Filippo Spinella

spinny.dev | Spinny03 | filippo-spinella | fili.spin2003@gmail.com | +39 351 980 5291

Experience

Wikicasa.it 2023 - Present
Back-end developer, Part-Time Genoa (Hybrid), Italy

- Improved the speed of reading agency data from MySQL to ElasticSearch by -83%
- Implemented a new microservice for managing geospatial polygons, going from ∼160ms to ∼13ms
- Migrated queries from MongoDB to ElasticSearch
- Pioneered the company's first Kubernetes cluster on bare-metal servers, managing the entire CI/CD pipeline with Bitbucket and ArgoCD.
- Created a microservice in Go and gRPC for converting images into different formats and sizes, handling over 25TB of data and 270 million images. Optimized storage on GCS Bucket, with a 40% cost reduction.
- Resolved major cache inefficiencies that slowed down application startup. Implemented a cache serialization system, reducing startup time by 45 seconds.
- Developed a RAG (Retrieval-Augmented Generation) system integrated with Wikicasa for the "TroverAI" feature: enables natural-language property search using Python, LangChain and the OpenAI APIs.

Ikinova 2022 - 2023
Full Stack developer, Part-Time Acqui Terme (Remote), Italy

- Restructured the backend from PHP to TypeScript (NodeJS), clustering it with PM2 for improved performance and scalability.
- Built a CI/CD environment from scratch using Mocha.js and GitHub Actions.
- Created a daemon that auto-synchronises patient data from an API of the Piemonte region
- Collaborated on developing front-end components using Ionic and Angular.

Education

University of Genoa 2025 - 2027
Masters of Science in Computer Science

University of Genoa 2022 - 2025
Bachelors of Science in Computer Science

CyberChallenge 2023 – 2023
National project for CyberSecurity training

Projects

University ChatBot 2024
Google x IANUA

- As part of a 4-person team, developed a chatbot using Next.js and Python to assist university students. The chatbot leverages Gemini for Retrieval-Augmented Generation (RAG) from the university's website.

Management Orders 2023
PLC company

- Led the development of an order management application for a PLC company, serving as both Project Manager and Back-end Developer.

Achievements

Participant - Samsung IC	The Samsung Innovation Campus is a specialization course in the IOT and innovation field.	2023
Student - IANUA	IANUA offers a special study plan to the most brilliant students at the University of Genoa (top 10th percentile).	2022
Mentee - LeadTheFuture	Among the few Italian students selected to be mentees for LeadTheFuture, a leading mentorship non-profit organization for students in STEM, with acceptance rate below 20%.	2022
Finalist - Kangourou	Kangourou is a nationwide math and programming competition for the best students in Italy	2022